Probing the Failure Boundary of Metacognitive Calibration in Large Language Models

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Abstract

TODO: Write abstract last.

1 Introduction

TODO: Write introduction after Section 3 is solid.

2 Background and Related Work

2.1 Calibration in Machine Learning

TODO:

2.2 Metacognition in Large Language Models

TODO

2.3 The Gap This Work Fills

TODO

3 Benchmark Design

Metacognition is the ability to know one's own knowledge base and reason on it. By virtue of it, a system can tell a lot- whether it can carry out a certain task, whether it can construct the solution of a task from the tasks that it knows, whether it has enough information to provably resolve an issue and many more.

This still

3.1 Construct Definition

Few attempts have been made for an operational definition of metacongition.

We attempt to formalize. Following are the routes we could have taken

1. **Second-Order Logic**: It is the closest formalism that directly correlates to these lines. It enables us to make statements on models. So hopefully it has something for us.

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2. **MetaProgramming**: Programming languages have consistently tried to achieve this abilities through reflection, templating, etc. Earliest attempts have been that of LISP and 3-LISP. We have failed, but in absolute terms. We still use reflection to useful extents. This can formalize.
3. **Semantic Understanding of submodules:** This is the most abstract one. This lies at intersection of mechanistic interpretability and Reflection. This will encounter us the Godel Incompleteness that Second Order Logic brings,

TODO: This is the most important paragraph in the paper. Write it first.

3.2 Task Regimes

3.2.1 Regime 1: Complete Information

TODO

3.2.2 Regime 2: Withheld Information

TODO

3.2.3 Regime 3: Misleading Information

TODO

3.3 Confidence Elicitation Protocol

TODO

3.4 Task Generation and Contamination Control

TODO

4 Scoring Methodology

Expected Calibration Error. **TODO: Add ECE definition with equation.**

Brier Score. **TODO**

Regime-Stratified Scoring. **TODO**

Null Baselines. **TODO**

5 Experiments

5.1 Models Evaluated

TODO

5.2 Aggregate Calibration Results

5.3 Regime-Stratified Results

5.4 Qualitative Analysis

TODO: Pick the most striking failure case from your runs and show it verbatim.

Table 1: Aggregate calibration results across models. Lower ECE and Brier score is better. Bold = best.

Model	ECE ↓	Brier Score ↓
Random baseline	–	–
Always-50%	–	–
GPT-4o	–	–
Claude 3.5 Sonnet	–	–
Gemini 1.5 Pro	–	–
Llama 3 70B	–	–

Table 2: ECE broken down by information regime. The gap between Regime 1 and Regime 3 reveals failure modes invisible to aggregate metrics.

Model	Regime 1 (Complete)	Regime 2 (Withheld)	Regime 3 (Misleading)
GPT-4o	–	–	–
Claude 3.5 Sonnet	–	–	–
Gemini 1.5 Pro	–	–	–
Llama 3 70B	–	–	–

6 Discussion

TODO

7 Limitations

TODO

8 Conclusion

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References